

Kubernetes: A Portable and Scalable Orchestration Layer for Container Applications

Kubernetes is a production grade open source orchestration system used to deploy, scale and manage containerised applications. It is based on the same principles that allow Google to run billions of containers a week. Read on to know more.



Docker has fundamentally changed the way we build, ship and run our applications. Docker has been a big winner when it comes to portability, primarily due to the fact that it is able to package an application and all its dependencies into a format that is easy to build, distribute and run. The availability of Docker runtime on multiple operating systems and the widespread support for it, across both cloud infrastructure providers and operating system vendors, has finally made it easier to achieve portability. This functionality has been a true enabler of DevOps, where the developers and the operations team can do a smooth handover.

So, while Docker with its easy-to-use tool set has made it convenient to launch a few containers on a single host, at best, the tooling is good enough for a few host machines running several containers on it. But when it comes to running large sites, one would probably need several large clusters, running thousands of nodes, if not more. This is not a trivial software problem. In fact, it lies in the realm of distributed software, which makes it a challenge to run Docker containers at scale. The challenge lies in the areas of managing the installation, rolling out upgrades, monitoring the Docker hosts and containers for their health, scaling up/down the number of containers needed as per the load and more.

From their perspective, it is important that developers do not have to worry about infrastructure and provisioning, even if the applications that they have *dockerised* need to scale. Similarly, from an operations point of view and, especially from a data centre angle, the days of allocating VMs and running your applications on them are gone. Infrastructure providers cannot run their data centres efficiently by simply spinning up VMs, which may end up not getting fully utilised.

So what we need is orchestration software that provides automated container deployment, scaling and management. In a nutshell, this is what Kubernetes is all about.

A bit of history

Google has been running some of the largest Web properties on the planet. Think of Gmail, Google Docs and you can imagine the millions of users at any given time. To support systems of this nature, Google has spent the last 10-15 years building an advanced infrastructure across the world. Before Docker was unleashed on the world, Google already understood the concept of containers, and has been running a distributed system that allowed the company to service its application requests by running and bringing down containers on demand. This inhouse system in Google was called Borg. It is reported